

UNIT 3

NUMBER SYSTEM



NUMBER SYSTEM

The **number system** is used for **reading, writing, counting, and calculation**. It is based on some characters called digits. Each number is made up of these digits.

3.1.1 Recognize the Base of a Number System

The number of digits a system uses is called its **base**. The present system of counting and computation is the base ten number system. This is called a **decimal number system**.

The basic numerals of the digits of the base ten system are ten in number and are as follows:

0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

The numbers from 0 to 9 are one-digit numbers. The numbers greater than 9 and less than 100 are **two-digit numbers**.

Similarly, the numbers greater than 999 and less than 10,000 consist of **four digits** and so on.

3.1.2 Define Number System with Bases 2, 5, 8, and 10**(i) Number System with Base 2**

A number system formed by two digits, 0 and 1, is known as a number system with base 2. It is also called a **binary number system**. Binary number system is not usually used in our daily system. But it is an important number system because it is used in every type of computer. Computers store data (information) in the form of 0s and 1s. Thus, binary system of numbers is of primary importance in our daily life which is the modern age of computers.

In binary system of numbers, 0 is the lowest numeral whereas 1 is the greatest numeral. In this system, zero is represented as 0_2 , one is represented as 1, and 2 is represented as 10_2 , read as one-zero to the base 2 and three is represented as 11_2 , read as one-one to the base 2.

(ii) Number System with Base 5

In this number system, the digits 0, 1, 2, 3 and 4 are involved. Here 0 is the lowest numeral and 4 is the greatest.

Zero is represented as 0_5 ,

One is represented as 1_5 ,

Two is represented as 2_5 ,

Three is represented as 3_5 ,

Four is represented as 4_5 ,

Five is represented as 10_5 and is read as one-zero to the base 5.

(iii) Number System with Base 8

In this system, the digits 0, 1, 2, 3, 4, 5, 6 and 7 are involved. Number system with base 8 is also known as **Octal System of Numbers**. In this system, the lowest numeral is zero and is represented as 0_8 . The greatest numeral in this system is seven and is represented as 7_8 . The number eight is represented as 10_8 , and nine is represented as 11_8 .

(iv) Number System with Base 10

In this system, the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 are involved. Number system with base 10 is also known as **Decimal Number System**. Decimal Number System is the most popular number system in the world because of its use in multiplication and division by 10 being easier.

Explain Number Systems with Bases 2, 5, 8, and 10

(i) Binary Number System (Number system with base 2)

In base 2 system, the smallest numeral is 0 and the greatest numeral is 1. In decimal system, the greatest numeral is 9, in the sense that when '1' is added to 9, it becomes 10 (read as ten). In the same sense, when '1' is added to '1' in the base two system, we get two, written as 10_2 and read as 'one-zero' and not ten.

Thus, in the binary system: One is written as 1_2 .

Now $1_2 + 1_2 =$ One plus one is Two, which is written as 10_2 .

Again, $10_2 + 1_2 =$ Two plus one is Three, which is written as 11_2 .

$11_2 + 1_2 =$ Three plus one is Four, which is written as 100_2 .

In the binary system, any number can be expressed as the sum of multiples of powers of 2.

For example:

$$13 = 8 + 4 + 1 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^0 = 111_2$$

$$27 = 16 + 8 + 2 + 1 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 11011_2$$

(ii) Number System with Base 5

The base 5 number system is based on the five fundamental digits 0, 1, 2, 3, and 4 to represent the numeric values.

In base 5 system, the smallest numeral is 0 and the greatest numeral is 4.

In decimal system $9+1=10$ but in base 5 system $4+1=10_5$ (Five, which is written as 10_5).

Similarly, $10_5 + 1_5 = 11_5$ (Six).

Now:

$$11_5 + 1_5 = 12_5 \text{ (Seven)}$$

$$13_5 + 1_5 = 14_5 \text{ (Nine)}$$

$$\text{Again } 14_5 + 1_5 = 20_5 \text{ (Ten) and } 20_5 + 1_5 = 21_5 \text{ (Eleven).}$$

Note: Any number in base 5 can be expressed as a sum of multiples of powers of 5.

For example:

$$(a) 87 = 75 + 10 + 2 = 3 \times 25 + 2 \times 5 + 1 \times 2 \quad \text{Or}$$

$$87 = 75 + 10 + 2 = 3 \times 5^2 + 2 \times 5^1 + 2 \times 5^0 = 322_5$$

(iii) Octal Number System (Number system with base 8)

The number system with base 8 consists of the following eight fundamental digits: 0, 1, 2, 3, 4, 5, 6, and 7 to represent the numeric values.

In base eight system, the smallest numeral is 0 and the greatest numeral is 7. In decimal system $9+1=10$ (Ten), but in octal number system $7_8 + 1_8 = 10_8$.

Similarly,

$$10_8 + 1_8 = 11_8 \text{ (Nine),}$$

$$12_8 + 1_8 = 13_8 \text{ (Eleven).}$$

$$11_8 + 1_8 = 12_8 \text{ (Ten),}$$

$$13_8 + 1_8 = 14_8 \text{ (Twelve)}$$

(iv) Decimal Number System (Number system with base 10)

The number system with base 10 is also called the decimal system.

We use this decimal number system in our daily life and business for counting and calculation.

It is based on ten digits (0, 1, 2, 3, 4, 5, 6, 7, 8, and 9).

EXERCISE 3.1

1. Answer the following:

(i) Define binary system of numbers. Write the first eight numbers in binary system.

(ii) Define number system with base 5. Write the first twelve numbers in base 5 system.

(iii) Define octal number system. Write the first sixteen numbers in this system.

(iv) Define decimal number system giving some examples.

2. Express the following decimal numbers as a sum of multiples of powers of 2:

(i) 11

(ii) 15

(iii) 21

(iv) 29

(v) 37

(vi) 51

(vii) 63

3. Express the following decimal numbers as a sum of multiples of powers of 5:

(i) 13

(ii) 24

(iii) 39

(iv) 47

(v) 59

(vi) 78

(vii) 89

(viii) 104

4. Express the following decimal numbers as a sum of multiples of powers of 8:

(i) 30

(ii) 51

(iii) 78

(iv) 98

(v) 114

(vi) 225

(vii) 340

5. Express the following decimal numbers as a sum of multiples of powers of 10:

(i) 201

(ii) 319

(iii) 4075

(iv) 56970

(v) 980762

CONVERSIONS

We have learnt the number systems with bases 2, 5, 8, and 10. These number systems are all place value number systems. The numbers used in the systems can be converted from one system to another system. The simple and the best method for conversion is the successive division method. With the help of this method, we can convert a number from the decimal system to another system.

Convert a number from decimal system to a system with base 2, 5, and 8 and vice versa:

(a) Conversion of numbers from Decimal System to a system with base 2, 5, and 8.

(i) Conversion of Numbers from Decimal System to base Two system.

Example 1. Convert 19 into binary number system.

Solution: We use successive Division Method.

In this method, the given number 19 is repeatedly divided by 2, till the remainder is 1.

Thus, continuing with the division of 19 by 2 and placing the remainders of each division on the right, unless the dividend is 1.

2	19
2	9 - 1
2	4 - 1
2	2 - 0
	1 - 0

Then arrange the remainder numbers, starting from bottom to get the required number in the binary system. Thus $19 = 10011_2$.

Example 2. Convert 276 into base two system.

Solution: We use successive division method.

2	276
2	138 - 0
2	69 - 0
2	34 - 1
2	17 - 0
2	8 - 1
2	4 - 0
2	2 - 0
	1 - 0

Explanation: Continue dividing 276 by 2. Place the remainder of each dividend on the right, unless the dividend is 1. Arrange the remainder numbers, starting from bottom to get the required number in base two system. Hence $276 = 100010100$.

(ii) Conversion of Numbers from Decimal system into Base Five system.

Example 1. Convert 678 into base five system.

Solution: We use Successive Division Method.

In this method the given number 678 is repeatedly divided by 5, till the dividend is 4 or less than 4.

Thus continue with the division of 678 by 5. Place remainder of each division on the right, unless the dividend is 4 or less than 4.

5	678
5	135 - 3
5	27 - 0
5	5 - 1
	1 - 0

Thus, $678 = 10103_5$.

Example 2. Convert 6065 into base five system.

Solution: Thus, $6065 = 143230_5$.

5	6065
5	1213 - 0
5	242 - 3
5	48 - 2
5	9 - 3
	1 - 4

(iii) Conversion of Numbers from Decimal System into Octal Number System.

Example 1. Convert 728 into octal number system.

Solution: We use Successive Division Method.

In this method, the given number 728 is repeatedly divided by 8, till the dividend is 7 or less than 7.

Now continue the division of 728 by 8. Place the remainder of each dividend on the right, unless the dividend is 1 or less than 7. Arrange the remainder numbers, starting from bottom to get the required number in base eight system.

Thus, $728 = 1330_8$.

8	728
8	91 - 0
8	11 - 3
	1 - 3

Example 2. Convert 2064 into octal system of numbers.

Solution: Thus, $2064 = 4020_8$.

8	2064
8	258 - 0
8	32 - 2
	4 - 0

EXERCISE 3.2

A. Convert the following decimal numbers into binary system:

- | | | | | |
|-----------|------------|-------------|------------|-----------|
| (i) 36 | (ii) 77 | (iii) 89 | (iv) 156 | (v) 580 |
| (vi) 407 | (vii) 654 | (viii) 786 | (ix) 995 | (x) 1200 |
| (xi) 1705 | (xii) 1830 | (xiii) 1096 | (xiv) 1291 | (xv) 1600 |

B. Convert the following decimal numbers into base five system:

- | | | | | |
|-----------|------------|------------|----------|----------|
| (i) 74 | (ii) 67 | (iii) 98 | (iv) 169 | (v) 5005 |
| (vi) 12 | (vii) 33 | (viii) 201 | (ix) 370 | (x) 5005 |
| (xi) 5050 | (xii) 5016 | | | |

C. Convert the following decimal numbers into octal system.

- | | | | | |
|------------|-------------|-------------|-------------|----------|
| i. 504 | ii. 1765 | iii. 20093 | iv. 33661 | v. 48360 |
| vi. 50607 | vii. 61278 | viii. 79118 | ix. 80877 | x. 99909 |
| xi. 119000 | xii. 129090 | xiii. 98765 | xiv. 198065 | |

(b) Conversion from other Number Systems to Decimal Number System:**(i) Conversion from Binary System to Decimal System:**

Conversion of a number from binary system to decimal system is explained with the help of following examples:

Example 1. Convert

(a) 11_2

(b) 111_2

(c) 110101_2 , into decimal system

Solution:

$$\text{(a)} \quad 11_2 = 1 \times 10_2^1 + 1 \times 10_2^0 = 1 \times 2^1 + 1 \times 2^0 = 2 + 1 = 3$$

$$\begin{aligned} \text{(b)} \quad 111_2 &= 1 \times 10_2^2 + 1 \times 10_2^1 + 1 \times 10_2^0 \\ &= 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= 1 \times 4 + 1 \times 2 + 1 \times 1 \\ &= 4 + 2 + 1 = 7 \end{aligned}$$

Hence, $111_2 = 7$.

(ii) Conversion from Five Base System to Decimal Number System

Example 1. Convert 42_5 into decimal form.

Solution:

$$\begin{aligned} 42_5 &= 4 \times 5^1 + 2 \times 5^0 \\ &= 4 \times 5 + 2 \times 1 \\ &= 20 + 2 \\ &= 22. \end{aligned}$$

Example 2. Convert 234_5 into decimal form.

Solution:

$$\begin{aligned} 234_5 &= 2 \times 5^2 + 3 \times 5^1 + 4 \times 5^0 \\ &= 2 \times 25 + 3 \times 5 + 4 \times 1 \\ &= 50 + 15 + 4 \\ &= 69. \end{aligned}$$

(iii) Conversion of numbers from octal number system into decimal number system

Example 1. Convert 506_8 into its decimal form.

Solution:

$$\begin{aligned} 506_8 &= 5 \times 8^2 + 0 \times 8^1 + 6 \times 8^0 \\ &= 5 \times 64 + 0 \times 8 + 6 \times 1 \\ &= 320 + 0 + 6 \\ &= 326 \end{aligned}$$

Thus, $506_8 = 326$.

EXERCISE 3.3

A. Convert the following binary numbers into decimal numbers.

1. 1101_2
2. 11011_2
3. 10111_2
4. 110101_2
5. 101001_2
6. 111101_2

B. Convert the following five base numbers into decimal number system.

1. 12_5
2. 30_5
3. 40_5
4. 12_5
5. 22_5
6. 131_5

C. Convert the following octal numbers into decimal numbers.

1. 47_8
2. 76_8
3. 40_8
4. 60_8
5. 70_8
6. 124_8

Add, Subtract, and Multiply Two Numbers with Bases 2, 5, and 8 Binary System (Numbers with Base 2)

(i) Addition of Numbers in Binary System

Addition in the binary system is very simple because only three pairs of numbers are formed for the addition.

$$0_2 + 0_2 = 0_2$$

$$0_2 + 1_2 = 1_2 + 0_2 = 1_2$$

$$1_2 + 1_2 = 10_2$$

Note:

The following result is obtained if 1_2 is to be added thrice.

$$1_2 + 1_2 + 1_2 = 11_2$$

$$\text{Then, } 1_2 + 1_2 + 1_2 = 11_2 + 1_2 + 1_2 = 11_2.$$

Example 1. Find the sum of 111_2 and 101_2 .

Solution:

First, write the given numbers according to the place value of the digits below each other.

Starting from the right, add 1 and 1 to get the sum 2, which is equivalent to 10 in binary system.

Write 0 and carry 1 to add it to 1, and 0, the result is again 10.

Write 0 and carry 1, add it to 1 and 1 to get 3 which in binary system is 11.

$$\text{Hence, } 111_2 + 101_2 = 1100_2$$

$$\begin{array}{r} 111_2 \\ + 101_2 \\ \hline 1100_2 \end{array}$$

Example 2. Add 1010_2 and 1111_2 .

Solution:

Thus, $1010_2 + 1111_2 = 11001_2$

$$\begin{array}{r} 1010_2 \\ + 1111_2 \\ \hline 11001_2 \end{array}$$

Subtraction of Numbers in Binary System:

The operation of subtraction in the binary system is just like addition.

Example 1: Subtract:

(a) 10_2 from 101_2

Solution:

Thus, $101_2 - 10_2 = 11_2$.

$$\begin{array}{r} 101_2 \\ - 10_2 \\ \hline \end{array} \quad \text{or} \quad \begin{array}{r} 101_2 \\ - 10_2 \\ \hline 11_2 \end{array}$$

(iii) Multiplication of Numbers in Binary System

The operation of multiplication in the binary system is simpler than addition; only care of position is to be taken. In multiplication, three pairs of numbers are obtained as shown below:

Table of Multiplication:

$$0 \times 0 = 0$$

$$0 \times 1 = 1 \times 0 = 0$$

$$1 \times 1 = 1$$

Example 1: Multiply 1101_2 and 101_2 .

Solution:

Here is the process

Hence,

$$1101_2 \times 101_2 = 1000001_2.$$

$$\begin{array}{r} 1101_2 \\ \times 101_2 \\ \hline 1101 \\ 00000 \\ 110100 \\ \hline 1000001_2 \end{array}$$

EXERCISE 3.4

A. Add the following in the binary number system:

1. $110_2 + 11_2$

2. $1010_2 + 110_2$

3. $1110_2 + 111_2$

4. $11011_2 + 11001_2$

5. $10101_2 + 110_2$

6. $111_2 + 11_2$

7. $1101_2 + 110_2 + 1001_2$

8. $1101_2 + 1001_2 + 11_2$

B. Solve the following in the binary number system:

1. $11101_2 - 11_2$

2. $111_2 - 101_2$

3. $1110_2 - 110_2$

4. $11111_2 - 1010_2$

5. $110111_2 - 1101_2$

6. $10011_2 - 101_2$

C. Simplify the following:

1. $(1111_2 + 101_2) - (1100_2 - 101_2)$
2. $(1010_2 + 10010_2) - (10101_2 - 1010_2)$
3. $11111_2 + 1010_2 + 101_2 - (10101_2 - 10010_2) - (10001_2 - 100_2)$

D. Simplify:

1. $101_2 \times 11_2$
2. $110_2 \times 101_2$
3. $1111_2 \times 100_2$
4. $110_2 \times 110_2$
5. $101_2 \times 1101_2$
6. $1010_2 \times 1010_2$

Base Five System

(i) Addition and Subtraction of Numbers in Base Five System

We know that the numerals 0, 1, 2, 3, 4 are used in the base five system. The method of addition and subtraction is explained by the following examples:

Example 1: Solve $4_5 + 1_5$.

Solution: The sum of four and one is five.

But five in the base five system is denoted by 10_5 .

Hence, $4_5 + 1_5 = 10_5$.

$$\begin{array}{r} 4_5 \\ + 1_5 \\ \hline 10_5 \end{array}$$

Example 2: Solve $3_5 + 4_5$

Solution: The sum of 3 and 4 is 7, which is written as "12" in the base-five system.

Thus,

$$3_5 + 4_5 = 12_5$$

$$\begin{array}{r} 3_5 \\ + 4_5 \\ \hline 12_5 \end{array}$$

Example 3: Add 123_5 and 312_5

Solution:

Thus,

$$123_5 + 312_5 = 440_5$$

$$\begin{array}{r} 123_5 \\ + 312_5 \\ \hline 440_5 \end{array}$$

Example 4: Find the sum of 4141_5 and 3421_5

Solution:

Thus,

$$4141_5 + 3421_5 = 13112_5$$

$$\begin{array}{r} 4141_5 \\ + 3421_5 \\ \hline 13112_5 \end{array}$$

Example 5: Subtract 231_5 from 424_5

Solution:

Thus,

$$424_5 - 231_5 = 143_5$$

$$\begin{array}{r} 424_5 \\ - 231_5 \\ \hline 143_5 \end{array}$$

Explanation:

- We get 3 after subtracting 1 from 4.
- We cannot subtract 3 from 2, so we borrow 1 from the leftmost 4 (leaving 3).
- This converts 2 into 7 (in base-5).
- $7 - 3 = 4$, and $3 - 2 = 1$.

Example 6: Subtract 4141_5 from 13012_5

Solution:

Thus,

$$13012_5 - 4141_5 = 3321_5$$

$$\begin{array}{r} 13012_5 \\ - 4141_5 \\ \hline 3321_5 \end{array}$$

Addition Table for Base-5 System

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	10
2	2	3	4	10	11
3	3	4	10	11	12
4	4	10	11	12	13

Summary

(i) The examples demonstrate addition and subtraction in Base-5, using the addition table for reference. The methods follow the same principles as decimal arithmetic, but adjustments are made based on the base-5 system.

(ii) Multiplication of Numbers in Base Five System

In fact, multiplying a number is the same as repeated addition. We adopt the same sense (order) under the operation of multiplication of numbers in the base-five system. This is the table of multiplication numbers in the base-five system.

Table of Multiplication for Base 5 System

x	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	11	13
3	0	3	11	14	22
4	0	4	13	22	31

Examples:

$$2_5 \times 2_5 = 4_5$$

$$3_5 \times 3_5 = 14_5$$

$$2_5 \times 3_5 = 11_5$$

$$3_5 \times 4_5 = 22_5$$

$$4_5 \times 4_5 = 31_5$$

$$10_5 \times 4_5 = 40_5$$

Examples 1: Find the product of 231_5 and 34_5 .

Solution:

Hence,

$$231_5 \times 34_5 = 20004_5$$

$$\begin{array}{r} 231_5 \\ \times 34_5 \\ \hline 2024 \\ + 12430 \\ \hline 20004_5 \end{array}$$

Examples 2: Find the product of 4321_5 and 123_5 .

Solution:

Hence,

$$4321_5 \times 123_5 = 1203033_5$$

$$\begin{array}{r} 4321_5 \\ \times 123_5 \\ \hline 24013 \\ 141420 \\ + 432100 \\ \hline 1203033_5 \end{array}$$

EXERCISE 3.5

A. Add the following:

1. $432_5 + 134_5$

4. $34321_5 + 24113_5$

2. $4333_5 + 343_5$

5. $43_5 + 154_5$

3. $435 + 145_5$

6. $1432_5 + 2341_5$

B. Solve the following.

1. $342_5 - 12_5$

4. $1203_5 - 134_5$

2. $202_5 - 14_5$

5. $4321_5 - 1234_5$

3. $3214_5 - 1403_5$

6. $4231_5 - 2304_5$

C. Solve the following:

1. $134_5 + 444_5 - 341_5$

4. $44444_5 - 321_5 - 444_5$

2. $1234_5 + 4321_5 - 444_5$

5. $22322_5 - 2111_5 - 232_5$

3. $1321_5 + 312_5 - 444_5$

D. Find the products of the following:

1. $423_5 \times 321_5$

4. $3231_5 \times 443_5$

2. $14_5 \times 13_5$

5. $233_5 \times 343_5$

3. $1134_5 \times 223_5$

6. $403_5 \times 243_5$

C. Octal number system (Numbers with base 8 system)

(i) Addition in Octal Numbers System:

We know that the numerals **0, 1, 2, 3, 4, 5, 6, and 7** are used in the base-eight system.

While adding, if the sum of two or more digits is greater than **7**, we divide the sum by **8**, then write the **remainder** and carry the **quotient** to the next digit.

Example 1: Add the following octal numbers.

(i) 5_8 and 7_8

Solution

(i) $5_8 + 7_8 =$

Thus,
 $5_8 + 7_8 = 14_8$

$$\begin{array}{r} 5_8 \\ + 7_8 \\ \hline 14_8 \end{array}$$

(ii) 64_8 and 74_8

(ii) $64_8 + 74_8 =$

Thus,
 $64_8 + 74_8 = 160_8$

$$\begin{array}{r} 64_8 \\ + 74_8 \\ \hline 160_8 \end{array}$$

The addition table will be helpful in finding the sums in the number system with base 8.

Example 2: Solve

$$445_8 + 443_8 = \boxed{}$$

Solution:

$$\begin{array}{r} 445_8 \\ + 443_8 \\ \hline 1110_8 \end{array}$$

Thus, $445_8 + 443_8 = 1110_8$

$$\begin{array}{r} 675_8 \\ + 756_8 \\ \hline 1653_8 \end{array}$$

Table of Addition in Octal System

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	10
2	2	3	4	5	6	7	10	11
3	3	4	5	6	7	10	11	12
4	4	5	6	7	10	11	12	13
5	5	6	7	10	11	12	13	14
6	6	7	10	11	12	13	14	15
7	7	10	11	12	13	14	15	16

(ii) Subtraction:

In subtracting one number from another with base 8, if the digit to be subtracted is **greater** than the corresponding digit, we **borrow 1** from the next column, which is equal to **8**. We **add 8** to this number, then subtract.

Example 1: Evaluate the following.

(i) $14_8 - 6_8$

Solution

$$(i) \ 14_8 - 6_8 = \boxed{}$$

Thus,

$$14_8 - 6_8 = 6_8$$

$$\begin{array}{r} 14_8 \\ - 6_8 \\ \hline 6_8 \end{array}$$

Example 1: Subtract:

(i) 2547_8 from 3756_8

Solution: Subtract 2547_8 from 3756_8 . Write $3756_8 - 2547_8$ in the vertical form. Then subtract.

Thus,

$$3756_8 - 2547_8 = 1207_8$$

$$\begin{array}{r} 3756_8 \\ - 2547_8 \\ \hline 1207_8 \end{array}$$

(ii) $604_8 - 247_8$

$$(ii) \ 604_8 - 247_8 = \boxed{}$$

Thus,

$$604_8 - 247_8 = 335_8$$

$$\begin{array}{r} 604_8 \\ - 247_8 \\ \hline 335_8 \end{array}$$

(ii) 476053_8 from 567442_8

Solution: Subtract 476053_8 from 567442_8 . Write $567442_8 - 476053_8$ in the vertical form. Then subtract.

Thus,

$$567442_8 - 476053_8 = 71367_8$$

$$\begin{array}{r} 567442_8 \\ - 476053_8 \\ \hline 71367_8 \end{array}$$

(iii) Multiplication in Octal Number System

This multiplication table in the octal system will help in solving products.

Example 1: Solve: $36_8 \times 43_8$

Solution

$$36_8 \times 43_8 = \boxed{}$$

$$\begin{array}{r} 36_8 \\ \times 43_8 \\ \hline 132_8 \\ + 1700_8 \\ \hline 2032_8 \end{array}$$

Thus, $36_8 \times 43_8 = 2032_8$

Multiplication in Octal Number System Table

+	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	10	12	14	16
3	0	3	6	11	14	17	22	25
4	0	4	10	14	20	24	30	34
5	0	5	12	17	24	31	36	43
6	0	6	14	22	30	36	44	52
7	0	7	16	25	34	43	52	61

Example 1: Evaluate the following.

(i) $446_8 \times 213_8$

Solution

(i) Writing $446_8 \times 213_8$ in vertical form.

$$\begin{array}{r} 446_8 \\ \times 213_8 \\ \hline 1562 \\ 4460 \\ + 111400 \\ \hline 117642_8 \end{array}$$

Thus, $446_8 \times 213_8 = 117642_8$

(ii) $30456_8 \times 234_8$

(ii) Writing $(30456)_8 \times (234)_8$ in vertical form.

$$\begin{array}{r} 30456_8 \\ \times 234_8 \\ \hline 142270 \\ 1116120 \\ + 6113400 \\ \hline 7374010_8 \end{array}$$

Thus, $30456_8 \times 234_8 = 7374010_8$

EXERCISE 3.6

1. Add the following:

(i) $275_8 + 52_8$

(iii) $4602_8 + 55741_8$

(v) $742351_8 + 35211_8$

(ii) $6245_8 + 7312_8$

(iv) $65234_8 + 51420_8$

2. Evaluate the following:

(i) $73_8 - 46_8$

(iii) $4326_8 - 3754_8$

(v) $163732_8 - 77766_8$

(ii) $200_8 - 173_8$

(iv) $76601_8 - 67706_8$

3. Evaluate the following:

(i) $67_8 \times 45_8$

(iii) $13470_8 \times 563_8$

(v) $763541_8 \times 4061_8$

(ii) $345_8 \times 171_8$

(iv) $30076_8 \times 324_8$

3. Addition, Subtraction, and Multiplication of Numbers with Different Bases

Let us consider the following examples.

Example 1:

Simplify $12_{10} + 12_5 + 110_2$ and express the answer in the binary system.

Solution: Evaluate $12_{10} + 12_5 + 110_2 = ?$

Let us convert 12_5 and 110_2 into decimal system and then simplify:

$$12_5 = 1 \times 5^1 + 2 \times 5^0 \\ = 1 \times 5 + 2 \times 1 = 5 + 2$$

or $12_5 = 7$

$$\text{and } 110_2 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ = 1 \times 4 + 1 \times 2 + 0 \times 1 = 4 + 2$$

or $110_2 = 6$

$$\text{Now } 12 + 12_5 + 110_2 = 12 + 7 + 6 = 25$$

On simplification, the result is 25.

Now converting the number 25 into binary system:

2	25
2	12 - 1
2	6 - 0
2	3 - 0
	1 - 1

or $25 = 11001_2$

Example 2:

Simplify $125 - 114_5 - 110_2$ and write the answer in base 5.

Solution: Let us convert 114_5 and 110_2 into decimal system and then simplify:

$$114_5 = 1 \times 5^2 + 1 \times 5^1 + 4 \times 5^0 \\ = 1 \times 25 + 1 \times 5 + 4 \times 1 = 25 + 5 + 4 = 34$$

$$110_2 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ = 1 \times 4 + 1 \times 2 + 0 \times 1 = 4 + 2 = 6$$

$$\text{Hence, } 125 - 114_5 - 110_2 = 125 - 34 - 6 \\ = 125 - 40 = 85$$

$$\text{Hence, } 125 - 114_5 - 110_2 = 85 = 320_5$$

Now converting the above result into a system with base 5.

5	85
5	17 - 0
	3 - 2

or $85 = 320_5$

Example 3:

Simplify $102_5 \times 1001_2$ and write the answer in base 5 and base 2.

Solution: $102_5 \times 1001_2 = ?$

Converting these numbers into decimal system,

$$102_5 = 1 \times 5^2 + 0 \times 5^1 + 2 \times 5^0 \\ = 1 \times 25 + 0 \times 5 + 2 \times 1 \\ = 25 + 2 = 27$$

$$\begin{aligned}
 1001_2 &= 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\
 &= 1 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1 \\
 &= 8 + 1 = 9
 \end{aligned}$$

$$\text{Now } 102_5 \times 1001_2 = 27 \times 9 = 243$$

Converting 243 into base five system.
Hence, $243 = 1433_5$.

5	243
5	48 - 3
5	9 - 3
	1 - 4

or $243 = 1433_5$

Again converting 243 into binary system.

2	243
2	121 - 1
2	60 - 1
2	30 - 0
2	15 - 0
2	7 - 1
2	3 - 1
	1 - 1

Therefore, $243 = 11110011_2$
Hence, $102_5 \times 1001_2 = 243$
 $= 1433_5 = 11110011_2$

EXERCISE 3.7

1. Simplify the following and express into the binary system:

(i) $27_8 + 14_{10} + 35_5$

(ii) $72_5 + 42_{10} + 110_8$

(iii) $83_{10} + 134_{10} + 1011_2 + 17_8$

(iv) $45 + 21_2 + 111_2 + 19_5$

(v) $38 + 52_{10} + 1001_2 + 27_{10}$

2. Simplify the following and express into a system with base 5:

(i) $41 - 24_{10} - 110_{10}$

(ii) $72 - 42_5 - 101_2$

(iii) $88_8 - 132_{10} - 110_2 - 12_5$

(iv) $201 - 145_5 - 102_{10}$

(v) $1300 - 1211_5 - 101_0 - 111_2$

3. Simplify the following and write into base five and binary systems:

(i) $21 \times 101_0$

(ii) $42 \times 101_2 \times 111_{10}$

(iii) $67 \times 110_{10} \times 101_8$

(iv) $34 \times 21_5 \times 110_2 \times 112_8$

(v) $81 \times 110_2 \times 101_2 \times 110_2$